AYUSH JHA

AI/ML Engineer | Machine Learning Engineer | Backend-focused ML Engineer | Applied AI Engineer

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SUMMARY

Applied AI/Machine Learning Engineer with experience building production-grade ML systems, LLM-powered applications, and backend-heavy AI platforms. Skilled in designing and implementing end-to-end machine learning pipelines, RAG-based systems, and deep learning models, including custom GPU-accelerated architectures.

Experienced in deploying real-time AI systems using FastAPI, Docker, and cloud infrastructure (AWS/GCP), with strong expertise in MLOps, inference optimization, and distributed systems. Focused on building high-performance, production-ready AI solutions for real-world applications.

SKILLS

Programming Languages: Python, C++
ML / AI: PyTorch, TensorFlow, Scikit-learn, CNNs, LSTMs, Transformers, LLMs, RAG, LangChain, LangGraph, Hugging Face, OpenAI API, Model Training, Fine-tuning, Inference Optimization, MLflow, LangSmith
Backend: Node.js, Python (Django, Fast API), NoSQL, MySQL, Redis, Vector DBs
Practices: ML Pipelines, Feature Engineering, Model Deployment, A/B Testing, Experiment Tracking, MLOps, Distributed Systems, WebSocket, REST APIs, TypeScript, Next.js
Familiar: Git, Github, Cloud (AWS, GCP), Kafka, Jenkins, Docker, Kubernetes, MlFlow, Hugging face, LLM, RAG

WORK EXPERIENCE

SORT-TREE. (Remote) London, UK
AI/ML Engineer (Contract) Nov 25 – Present

- Designed and deployed ML-driven analytics systems processing 50K+ campaign data records daily, enabling near real-time campaign optimization
- Built FastAPI-based backend services handling 100+ API requests/min to serve ML model predictions in production dashboards
- Engineered data pipelines integrating Meta/Facebook & Instagram Graph APIs, aggregating data across 20–30 ad accounts.
- Developed real-time dashboards tracking 10+ campaign KPIs, improving visibility for stakeholders and reducing manual reporting effortIntegrated ML outputs into scalable backend systems supporting concurrent campaign monitoring across multiple clients

SARANYU TECHNOLOGIES. Bengaluru, Karnataka
Software Engineer Sep 24 – Oct 25

- Developed backend APIs using Node.js (Fastify) and PHP supporting 10k+ concurrent users across multiple client applications
- Built frontend features using Next.js, contributing to improved page responsiveness from 60-90+ Lighthouse score and user experience across production systems
- Implemented CI/CD pipelines using GitHub Actions, automating deployments and reducing manual release effort across multiple environments
- Collaborated within Agile teams of 6–8 engineers, contributing to sprint delivery, code reviews, and maintaining code quality

APPKNOX. Bengaluru, Karnataka
Software Engineering Intern May 24 – Aug 24

- Added SARIF report generation to AppKnox CLI, enabling GHAS integration. Worked on GitHub Actions for CI/CD automation, ensuring seamless software delivery for 30+ clients with GHAS enabled.
- Developed and integrated a Jenkins CI/CD plugin, automating security scanning for 200+ enterprise clients.
- Engineered a Python-based web scraper to systematically collect and store metadata for 15,000+ Play Store apps, enhancing data processing efficiency by 35%.

PROJECTS

ZYNK – AI POWERED VOICE AND GESTURE CHROME EXTENTION.
Intelligent Browser Automation System (Computer Vision + Voice AI, Chrome Web Store)

- Built and deployed an AI-powered browser automation system using voice and gesture inputs with sub-100ms latency
- Deployed a production-grade backend on GCP (e2-micro VM, pm2 process management) with FastAPI, Supabase (PostgreSQL + pgBouncer connection pooling, SSL), in-memory Bearer token auth with silent refresh, rate limiting to control OpenAI API costs, and CORS-configured Vercel-deployed Next.js frontend.
- Enabled semantic understanding of voice commands, LLM, and LangSmith observability.
- Integrated faster-whisper with VAD for real-time speech recognition and command execution

DEEP SUPER RESOLUTION.

Custom GPU-Accelerated Image ResNet System (Deep Learning, CUDA, CuPy, C++, MLOps)

- Built a 2× Image Super-Resolution Residual CNN from scratch without PyTorch/TensorFlow; implemented custom forward/backward propagation, convolution layers, loss functions, and Adam optimizer using CuPy with PSNR/SSIM evaluation on DIV2K dataset.
- Engineered CUDA C++ kernels for convolution and matrix multiplication via PyBind11; benchmarked CPU vs GPU performance achieving significant 20-30x inference speedup through kernel fusion, memory optimization, and parallel execution on NVIDIA RTX 1650 GPU.
- Built end-to-end MLOps pipeline with Docker containerization, GitHub Actions CI/CD, MLflow experiment tracking, and drift monitoring covering data ingestion, training, evaluation, versioning, and deployment stages.
- Deployed a FastAPI + Next.js web application for real-time image super-resolution inference with GPU/CPU support, live monitoring of GPU utilization and latency, and comprehensive unit, integration, and gradient-check test coverage for over 50+ unit test files with pytest

ML MODELS.

ML Models from Scratch: - Production-Style ML Systems (Python, Fast API, Kafka, PostgreSQL)

- Built 3 end-to-end ML systems on structured datasets (50K–200K records each): **Marketing & Housing Price Prediction (Linear Regression)**, **Real-Time Fraud Detection (Logistic Regression)**, and **Loan Approval Decision System (Decision Tree)**.
- Designed end-to-end ML pipelines including PostgreSQL data ingestion, feature engineering with 10+ versions, custom gradient descent training, model persistence, and structured inference workflows without using pre-built ML libraries.
- Built production-style architecture using Fast API (concurrent APIs), Kafka (streaming) for over 20k Credit card fraud dataset, WebSocket (real-time output), Prometheus & Grafana (monitoring), and implemented retraining, logging, and model versioning with rollback support.

EDUCATION

REVA UNIVERSITY — Bengaluru, Karnataka

Bachelor of Engineering — Jul 2019 - Jul 2023

Major ECE; Minor Computer Science.
CGPA: 8.3 /10.

ADDITIONAL

Certifications & Training:

- Deep Learning Specialization(Deeplearning.AI)
- Machine Learning Specialization (Deeplearning.AI)
- Airtribe Backend Launchpad
- Tensor Flow Applications